FOR SEPARATED FLAT FINITELY PRESENTED MAPS THE FINITE PART IS ETALE, SEPARATED AND REPRESENTABLE

THE STACKS PROJECT AUTHORS

Editorial note. No omission markers in-span.

Allowed standard prerequisites (editorial).

- Diagonal construction. lemma-finite-diagonal (carded L26653). Supplies the closed complement E whose openness is now to be upgraded.
- Closedness of the parts. lemma-finite-separated (carded L26652). Z_1, Z_2 closed in T' x_Y X.
- Openness of flat finitely presented morphisms. Morphisms of Spaces lemmas base-change-finite-presentation, base-change-flat, fppf-open (cross-chapter, named block). Makes pr_0|_{Z_1} open so that E is open.
- Representability of locally quasi-finite separated morphisms. Morphisms of Spaces lemmas etale-locally-quasi-finite and locally-quasi-finite-separated-representable, proposition locally-quasi-finite-separated-over-scheme (cross-chapter, named block). Closes representable and the scheme case.

Source reference. Original equation 04PC: sources/stacks-spaces-more-groupoids/pinned-original-source.tex, source lines 1519–1519. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 04PK: sources/stacks-spaces-more-groupoids/pinned-original-source.tex, source lines 1744–1744. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 04PH: sources/stacks-spaces-more-groupoids/pinned-original-source.tex, source lines 1672–1672. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original proposition 04QH: sources/stacks-spaces-more-groupoids/pinned-original-source.tex, source lines 2012–2012. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Date: stacks-project:spaces-more-groupoids.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original reference 03XR: focused-seeds/geometry/S9-smg-finite-separated-flat-locally-finite-presentation/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 03MO: focused-seeds/geometry/S9-smg-finite-separated-flat-locally-finite-presentation/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 03XU: focused-seeds/geometry/S9-smg-finite-separated-flat-locally-finite-presentation/ref-tags.txt, source lines 7–7. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0464: focused-seeds/geometry/S9-smg-finite-separated-flat-locally-finite-presentation/ref-tags.txt, source lines 8–8. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 042S: focused-seeds/geometry/S9-smg-finite-separated-flat-locally-finite-presentation/ref-tags.txt, source lines 9–9. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0418: focused-seeds/geometry/S9-smg-finite-separated-flat-locally-finite-presentation/ref-tags.txt, source lines 10–10. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 03XX: focused-seeds/geometry/S9-smg-finite-separated-flat-locally-finite-presentation/ref-tags.txt, source lines 11–11. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let S be a scheme. Let $f : X \to Y$ be a morphism of algebraic spaces over S which is separated, flat, and locally of finite presentation. In this case*

(1) *$(X/Y)_{fin} \to Y$ is separated, representable, and étale, and*
(2) *if Y is a scheme, then $(X/Y)_{fin}$ is (representable by) a scheme.*

Proof. Since f is in particular separated and locally of finite type (see Morphisms of Spaces, Lemma 0464) we see that $(X/Y)_{fin}$ is an algebraic space by Proposition 04QH. To prove that $(X/Y)_{fin} \to Y$ is separated we have to show the following: Given a scheme T and two pairs (a, Z_1) and (a, Z_2) over T with identical first

component satisfying 04PC there is a closed subscheme $V \subset T$ with the following property: For any morphism of schemes $h : T' \to T$ we have

$$h \text{ factors through } V \Leftrightarrow \left(T' \times_T Z_1 = T' \times_T Z_2 \text{ as subspaces of } T' \times_Y X\right)$$

In the proof of Lemma 04PK we have seen that $V = T' \setminus E$ is an open subscheme of T' with closed complement

$$E = \mathrm{pr}_0|_{Z_1}(Z_1 \setminus Z_1 \cap Z_2)) \cup \mathrm{pr}_0|_{Z_2}(Z_2 \setminus Z_1 \cap Z_2)) .$$

Thus everything comes down to showing that E is also open. By Lemma 04PH we see that Z_1 and Z_2 are closed in $T' \times_Y X$. Hence $Z_1 \setminus Z_1 \cap Z_2$ is open in Z_1. As f is flat and locally of finite presentation, so is $\mathrm{pr}_0|_{Z_1}$. This is true as Z_1 is an open subspace of the base change $T' \times_Y X$, and Morphisms of Spaces, Lemmas 03XR and Lemmas 03MO. Hence $\mathrm{pr}_0|_{Z_1}$ is open, see Morphisms of Spaces, Lemma 042S. Thus $\mathrm{pr}_0|_{Z_1}(Z_1 \setminus Z_1 \cap Z_2))$ is open and it follows that E is open as desired.

We have already seen that $(X/Y)_{fin} \to Y$ is étale, see Proposition 04QH. Hence now we know it is locally quasi-finite (see Morphisms of Spaces, Lemma 03XU) and separated, hence representable by Morphisms of Spaces, Lemma 0418. The final assertion is clear (if you like you can use Morphisms of Spaces, Proposition 03XX). □